GMR angle error extension

XENSIV™ angle sensors

About this document

Scope and purpose

This document describes the magnetic field strength across various different temperature conditions in relation to angle deviation for Infineon's XENSIV™ angle sensors using giant magnetoresistance (GMR) sensor technology.

Intended audience

The intended audiences for this document are design engineers, technicians, and developers of electronic systems interested in integrating the XENSIV™ GMR-based angle sensors into their designs.

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1 Influences on iGMR angle sensor error

The specification of iGMR sensor components defines absolute angle error for certain defined operating conditions (magnetic field range, temperature). Typically, the sensor elements operate well for operating conditions outside the given range of the specification.

This application note quantifies the impact to the overall angle error of iGMR sensors, namely the TLE5009 and TLE5012B. All angle errors listed relate to the angle error including lifetime drift and temperature variation.

This application note describes the main contributors impacting iGMR angle error and explains how to calculate the resulting angle error if operating the devices under different magnetic field and temperature ranges.

Main contributors for:

- Increased magnetic fields in combination with elevated temperatures results in increased lifetime drift of the device. This angle error adder is not present at 0 hour operation but happens irreversibly over lifetime operation. On the other side, a reduced temperature range offers the potential to operate the iGMR sensor at higher magnetic fields without any impact on the specified angle error. In Section 2.4 these angle error adders are quantified.
- Operation of the iGMR sensor at lower magnetic fields than specified leads to an increased angle error due to anisotropy and hysteresis effects. This angle error is completely reversible, it appears if the magnetic field drops below the value given in the specification and it immediately disappears after re-entering the specified magnetic field range.

Both effects are cumulative and must be added. The calculation of the final resulting angle error is done in the following manner:

Specified angle error given the product datasheet (base angle error)
PLUS
Angle error adder due to increased magnetic fields (see Table 1)
PLUS
Angle error adder due to operation at lower magnetic fields (see Table 2)

Some exemplary calculations are provided in Section 2. In Section 2.4, you can find tables for each product showing the maximum angle error adder for all evaluated operating conditions. The “base angle error” must be taken from the datasheet of each product.

1.1 Angle error impact of increased magnetic fields as function of temperature

Increased magnetic fields in combination with elevated temperatures results in increased lifetime drift of the device. This angle error adder is not present at 0 hour operation but happens irreversibly over lifetime operation. On the other side, reduced temperature ranges offer the potential to operate the iGMR sensor at higher magnetic fields without any impact on the specified angle error.

The magnetic fields in the table below refer to the magnetic field applied at room temperature. The evaluation was done using magnets with a temperature coefficient of -0.13 %/K and assumes that the magnet faces the same temperature as the sensor.

Table 1 Angle error impact of increased magnetic fields as function of temperature

Max. operating temperature T_j	Max. magnetic field at 25°C	
	60 mT	70 mT
150°C	0.16°	0.32°
125°C	0.10°	0.21°
100°C	0°	0.10°
85°C	0°	0°

Valid for TLE5012B, TLE5009

1.2 Angle error impact due to operation at lower magnetic fields than specified

Operation of the iGMR sensor at lower magnetic fields than specified leads to an increased angle error due to anisotropy and hysteresis effects. This angle error is completely reversible, it appears if the magnetic field drops below the value given in the specification and it immediately disappears after re-entering the specified magnetic field range.

The magnetic fields in the table below refer to the magnetic field applied at room temperature. The evaluation was done using magnets with a temperature coefficient of -0.13%/K and assumes that the magnet faces the same temperature as the sensor.

Table 2 Angle error impact due to operation at lower magnetic fields than specified

Product	Adder if operating down to	
	25 mT	20 mT
TLE5012B	0.1°	0.2°
TLE5009	0°	0.1°

2 Calculation examples

This section shows some practical examples of how the angle error is modified when extending the magnetic range.

2.1 TLE5012B with 30-70 mT at -40°C to 150°C

The TLE5012B has a specified angle error of 1° over lifetime and temperature including autocalibration as far as the magnetic field is within the specified range of **30-50 mT**.

If the TLE5012B is used with an increased magnetic field range of **30-70 mT** it can be done, knowing that the angle error would be modified as shown in Table 1 and Equation 1.

$$1° + 0.32° = 1.32°$$

Equation 1 Angle error modification 50-70 mT

2.2 TLE5012B with 20-50 mT at -40°C to 150°C

If the TLE5012B is used with an increased magnetic field range of **20-50 mT** it can be done, knowing that the angle error would be modified as shown in Table 2, Equation 2 and Equation 3.

$$1° + 0.2° = 1.2°$$

Equation 2 Angle error modification 20-25 mT

$$1° + 0.1° = 1.1°$$

Equation 3 Angle error modification 25-30 mT

Now the worst case is taken, which would be Equation 2, 1.2° angle error.

2.3 TLE5012B with 20-70 mT at -40°C to 150°C

The TLE5012B has a specified angle error of 1° over lifetime and temperature including autocalibration as far as the magnetic field is within the specified range of 30-50 mT.

If the TLE5012B is used with an increased magnetic field range of **20-70 mT** it can be done, knowing that the angle error would be modified as shown in Table 1, Table 2, Equation 4, Equation 5, and Equation 6.

$$1° + 0.32° + 0.2° = 1.52°$$

Equation 4 Angle error modification 20-25 mT

$$1° + 0.32° + 0.1° = 1.42°$$

Equation 5 Angle error modification 25-30 mT

$$1° + 0.32° + 0.0° = 1.32°$$

Equation 6 Angle error modification 50-70 mT

Now the worst case is taken, which would be Equation 4, 1.52° angle error.

2.4 Angle error for different operating conditions

The following tables give the angle error adders at different operating conditions that comes on top to the values given in the specification.

Table 3 Angle error adder for different operating conditions for TLE5012B

	Max. operating temperature T_j			
	85°C	**100°C**	**125°C**	**150°C**
30-50 mT	0.00°	0.00°	0.00°	0.00°
30-60 mT	0.00°	0.00°	0.10°	0.16°
30-70 mT	0.00°	0.10°	0.21°	0.32°
25-50 mT	0.10°	0.10°	0.10°	0.10°
25-60 mT	0.10°	0.10°	0.20°	0.26°
25-70 mT	0.10°	0.20°	0.31°	0.42°
25-50 mT	0.20°	0.20°	0.20°	0.20°
25-60 mT	0.20°	0.20°	0.30°	0.36°
25-70 mT	0.20°	0.30°	0.41°	0.52°

Table 4 Angle error adder for different operating conditions for TLE5009

	Max. operating temperature T_j			
	85°C	**100°C**	**125°C**	**150°C**
25-50 mT	0.00°	0.00°	0.00°	0.00°
25-60 mT	0.00°	0.00°	0.10°	0.16°
25-70 mT	0.00°	0.10°	0.21°	0.32°
25-50 mT	0.10°	0.10°	0.10°	0.10°
25-60 mT	0.10°	0.10°	0.20°	0.26°
25-70 mT	0.10°	0.20°	0.31°	0.42°

References

[1] Infineon Technologies AG: *XENSIV™ TLE5009 EXXXX datasheet*; Available online

[2] Infineon Technologies AG: *XENSIV™ TLE5012B EXXXX datasheet*; Available online

Revision history

Document revision	Date	Description of changes
1.20	2011-08-03	Changed ‘accuracy’ term with “angle error” for better formulation
1.30	2025-09-29	Updated template Removed discontinued products

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